Hybridoma Lec(1)

* what we will take (labs):
1] immunization of mouse
2] screening test: testing the mouse's serum (Abs + Bcells)
booster dose
ELISA

(tissue culture ↓)
3] Myeloma cell (cancer cell) (cancerous plasma cell)
4] isolation of B-cell from the spleen of the mouse
very rich in B-cells
5] fussion of Myeloma + B cells (M/B → hybridoma cell)
6] tissue culture flask
7] cloning seperation of two hybridoma cells
8] large scale preperation
9] characterization

* Adjuvant

complete	incomplete
- oil + Mycobacterium tubercolosis	- only oil
- at 1st injection mostly IgM, few IgG	- at any time except 1st, 2nd mostly IgG

dose: for single mouse BSA, 50-100 μg BSA
BSA: 50-100 μL
BSA: 100 μL + CFA 100 μL = 200 μL = 0.2 μL

Hybridoma LAB

LAB (1). Solution

* solution (solute + solvent)

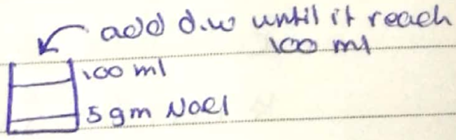
1. percentage solution (%)
2. motor solution (M)
3. normality solution (N)
4. buffer solution

* percentege solution has 3 types:

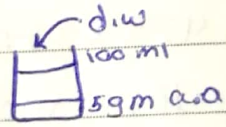
1. w/v % (weight per volume of solution :- ※ of gm solute per 100 ml solution

   solute → liquid (pure) → solid

   ex. 5 % w/v $NaCl_{(s)}$



ex. 5% w/v acetic acid$_{(L)}$ "pure form"



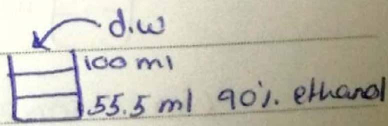
2. v/v % :- ※ of ml solute per 100 ml solution in pure form

   50% ethanol from 100% ethanol → pure

   50% ethanol from 90% ethanol → not pure

   in non-pure solutions we use this equation: $C_1V_1 = C_2V_2$

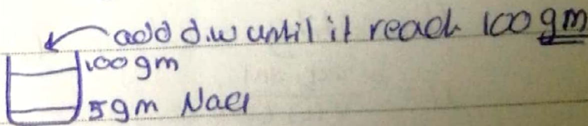
   $90 \cdot V_1 = 100 \cdot 50 \rightarrow V_1 = 55.5$ ml



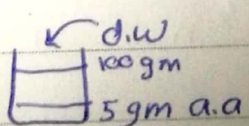
3. w/w % :- ※ of g solute per 100 g solution

   solute → liquid (pure) → solid

   ex. 5% $NaCl_{(s)}$ w/w



ex. 5% acetic acid$_{(L)}$



* Molar solution (M)

$M = \frac{\text{* moles (solute)}}{V_{(L)} \text{ (solution)}}$

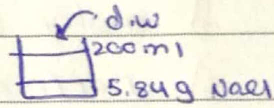
1] when solute is solid :

$\text{* moles} = \frac{\text{mass}}{\text{molar mass} \times V_{(L)}}$

ex. $NaCl_{(s)}$: m.m = 58.4 , M = 0.5 , V = 0.2 L , 200 mL

mass = M x m.m x V

= 0.5 x 58.4 x 0.2 = 5.84 g / 0.2 L



2] when solute is liquid :

~~[illegible]~~ $\text{* ml/1L} = \frac{M \times m.m}{\text{specific gravity} \times \text{assay / \% / purity}}$

stock

written as desimel ÷100

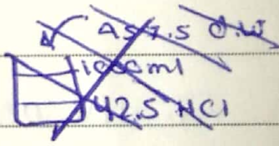
ex. HCl % = 36.5 %

m.m = 36.5 , S.G = 1.19 , M = 0.5

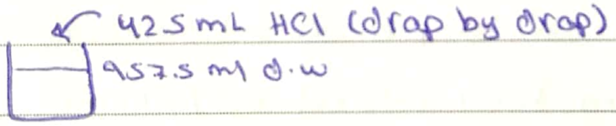
calculate * ml/1L , * ml/0.5 L ?

$\text{* ml/1L} = \frac{0.5 \times 36.5}{1.19 \times 0.365} = 42.5 \text{ mL HCl}$

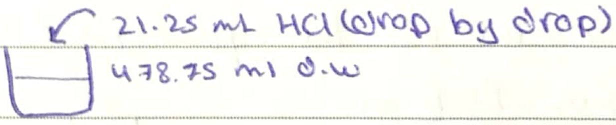
42.5 x 0.5 L = 21.25 mL HCl



1L : 1000 mL − 42.5 = 957.5 mL



0.5 L : 500 mL − 21.25 = 478.75 mL



in strong acids we add water then the acid

* Normality solution (N)

$N = \frac{\text{* eq. sol}}{V_{(L)} \text{ (solution)}}$

$N = \frac{\text{mass}}{m.m \times V} \times n$ n = | velence change |

= Mn

Hybridoma Lec (2) Immunization

* why do we use mice?
- first of all, we use young mice (8–12 weeks old)

bulb/c mice B + M bulb/c mice

- we use them to prevent rejection and they have a mature immune system and we need a good immune response
- sex doesn't matter
- mice:
  1) immunization 2) ascitic fluid 3) feeder layer

* Ag:
- most of Ags are soluble proteins
- proteins have high molecular weight
- proteins that are less than 1000 MW are considered non-atigenic (hapton – carrier)
- (CHO, lipids, N.A) ⤳ very weak Ags, very weak immune response
- N. acid (high dose) $\xrightarrow{\text{link to}}$ carrier
- dose of Ags: proteins → 10–100 μg
- injection root: IP (intraparotonial)
- volume of Ag = 200 μL (not the dose!)
  very low dose (10 μg) or very high dose (100 μg) will not produce immune response, it'll produce tolerance
- Ag could also be a cell ex. Ab against bacteria → bacterial cell, Ab against cancer cells → cancer cells

- injection roots:

SC (تحت الجلد)

IP (في البطن) ↝ if the volume is more than 200 ML it's okay

IV (في الوريد) rarely used because veins are really small

- we keep injecting until we get a positive result in screening (usually it takes 3 injections)

positive results is when Abs are present

* Haptens:
- small molecules that produce no immune response
- fatty acids are not immunogenic so we conjugate it with protein to become immunogenic
- Ag (↑i.r) + adjuvant (complete freund adjuvant CFA water in oil + micobacterium)
- $f.a \xrightarrow{\text{strong chemical bond}}$ protein (carrier)

cholestrol — protein (usually BSA / KLH)

fatty acid and cholestrol both are not immunogenic, when we conjugate them chemically with protein they become antigenic

* Adjuvant
- very important, without it we get a very weak immune response
- ~~doe~~ inhance immune response but doesn't change the epitope (insted of 20 i.r it becomes 60)
- kinds of adjuvant:

1] complete fruid adjuvant (CFA) water + oil + M.t (activate the innate imnunity)

2] incomplete fruid adjuvant (IFA) water + oil

• mechanism:

injection + CFA → oil (we should mix them vigourusly)

water soluble (protein)

Ag must be forgein, for example we can't inject a mouse with mouse protein!

•• mixing mechanism:

first method ① we put one drop of adjuvant and we put the cap then mix using vortex, then we put another drop and mix with vortex again

(we put the cap because it contain bacteria)

second method ② we put protein (Ag) in a syrenge and the adjuvant in another syrenge and connect them by a plastic (rubber) tubes and start to inject the protein toward the adjuvant, then inject them toward the Ag until they're mixed

(the oil is very thick, so if we inject hershly the tube may come off, so we prefer the first method)

• we must mix very well so that the Ag will be mixed with oil

when mixed very well a layer over H2O will form; other wise it will all be mixed with water

✱ testing:

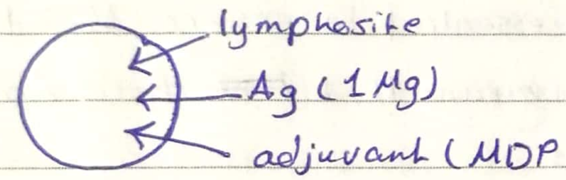
we cut the mouses' tail tip and take 2-3 drops of blood

1st injection $\xrightarrow{\text{week}}$ 2nd injection $\xrightarrow{\text{week}}$ 3rd injection

-ve -ve +ve

$\xrightarrow{\text{3-4 days}}$ very high immune response (in vivo immunization)

* for preperation of monocloned Ab for immunization the Ag shouldn't be pure
* in vitro immunization (in petridish)



Hybridoma LAB

LAB (2) Immunization

* immunization: injection for immune system by forgein adjuvent
* B.S.A

B.S.A 50-100 Mg (strong immun.) 1000 Mg / 1000 ML = 1 mg/ml

100 Mg = 0.1 ml

+ C.F.A (1st injection)

B.S.A : C.F.A ↪ 1:1

100 ML + 100 ML ↪ 200 ML / mouse = 0.2 ml

+ I.F.A (2nd injection)

B.S.A : I.F.A ↪ 1:1

100 ML + 100 ML ↪ 200 ML / mouse = 0.2 ml

* I.P ↪ handling

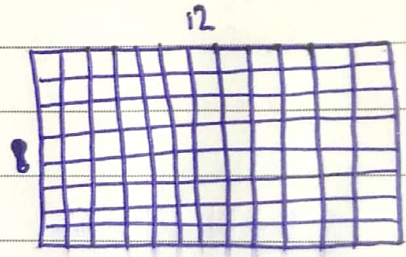
Hybridoma Lec (8)

* screening test (testing the serum)
testing the serum is essential to ensure that the serum contain Abs and this means that ~~that~~ these Abs are secreted by B-cells
specific Abs against your injected Ag, we know the antigen (BSA)

* different methods for testing the serum:

1] IF
2] RIA
3] ELISA (most common)

* ELISA test:
200 ml
wells' bottom:
flat or v-shape or u-shape

microtiter plate


→ Ag (1-10 μg)
Ag specify the type of the well

- counting the antigens (BSA)
link the Ag to the bottom of the well
- add the antibody (tested mouse serum Ab)
antibody will join specifically to the epitopes of Ag
- false-negative ⟶ when we take low concentration of Ag
- non-reproduceable results ⟶ when we take high concentration of Ag
- Ags are expensive because they must be as pure as possible for the ELISA test, other wise there will be contamination

- Ag might be protein, cells, bacteria, etc.
- procedure:

① we add blocking solution: a protein that prevent Abs from bonding with the walls of the well ex. BSA, gelatin
if we don't use the blocking solution we might get false-positive results

② we add the mouse serum (10 µL): if the serum has Ab against the antigen, the result will be a colored solution, other wise it will be colorless

③ washing: serum is not pure, so we wash 3 times to remove non-specific molecules
we wash using buffer (carbonate)

④ second Ab conjugate: this Ab is connected to an enzyme and is specific against the first Ab

- we make the second Ab by injecting the mouse Ab (first Ab) into a rabbit and taking the anti-mouse (second Ab)
- the first antibody is made by the following:

$$\text{serum} \xrightarrow{\text{ammonium sulfate } 33\%} \text{PP} + \text{Abs (mouse IgG)}$$

- after taking the antimouse we add the enzyme + gluteraldehyde (Ag – CHO – CHO – E)
the enzyme the we use is horseradish peroxidase (stable, cheap, comes from a plant)

⑤ wash

⑥ substrate (colorless): enzyme will break down the substrate making it a colored compound

- substrate (OPD + H_2O_2)
- color intensity is directly propotional to the concentration of the first Ab

⑦ in a second well, we use irrelavant Ag
in a third well, we use irrelavant Ab
in a forth one, we use buffer
- irrelevant Ag ⤳ if a color showed it means that there's a technical issue or the Ab isn't monoclonal
- irrelevent Ab ⤳ it's okay if a very light color showed
- buffer ⤳ instead of 2nd Ab, if a color showed there's a technical error (issue)
- standard positive control ⤳ if there's no color it means that there's a technical issue

* Alkaline phosphatase ⤳ removes the phosphate, once it's removed there's color

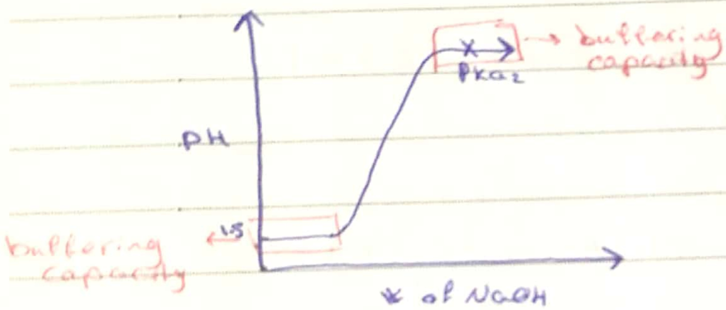
HYBRIDOMA LAB

LAB (3). BUFFER

* solution that contain weak acid and its conjugate base
or weak base and its conjugate acid
which resist change in pH during buffering capacity

* buffering capacity = pka ± 1 $HA \rightleftharpoons [A^-] + [H^+]$

P.B.S. PKa = 6.8 ⟶ 5.8 – 7.8



* how to prepare buffer:

1. ~~mixed~~ mix fixed volume of weak acid (weak base)
with ~~conjugate~~ fixed volume of conjugate base
(conj. acid)

disadvantage: ① not all buffer can be prepared by this way
② PH for D.W ≠ 7

2. mix fixed volume of weak acid (weak base) with unknown
volume of conjugate base (conjugate acid)

3. prepare weak acid and it conjugate base
↓PH ⟶ ↑1M NaOH
↑PH ⟶ ↓ 1M HCl (drop by drop)

* ~~pa~~ phosphate buffer saline (P.B.S)

PO_4 — acid + base — NaCl

0.1 M PH = 7.35 - 7.45

0.1 L ≃ 100 ml

used for: ① dilution of stock solution

② washing, remove previous substance

content: ① NaCl 0.8 g/0.1 L

② KCl 0.02 g/0.1 L

③ Na_2HPO_4 0.115 g/0.1 L → conjugate base

④ KH_2PO_4 0.02 g/0.1 L → weak acid

⑤ water

* prosedure:

1] weigthing salts (1+2+3+4)

2] take 90 ml D.W

3] mix salt with D.W in beaker on magnitic stirrle

4] adjust PH ≃ 7.35 - 7.45

5] complete volume up to 100 ml marker by D.W

6] put P.B.S in Durra bottels

label: P.B.S PH = 7.45 Hybridoma sec 2 20-1

7] sterilization by Autoclave: T = 121°c, time = 30 - 1 hr. pre. = 1

Hybridoma Lec (4) Tissue culture lab

* in tissue culture lab we work with cells, petridish, flask (in vitro)
* advantages of tissue culture (invitro)
1) can control the environment ex. control Ph, temperature, etc.
2) easy to deal with
3) quick
4) can control the concentration of each chemical
* finite and continuous tissue culture
1) finite: (limited) dealing with lymphocytes (B,T cells) can't keep lymphocytes in tissue culture forever (limited division)
2) continuous: (unlimited) dealing with cancer cells, they keep dividing as long as we provide them with nutrients in proper conditions, best example is celline
* the most important and dangerous thing that can happen to tissue culture is contamination! so the main thing that we should concentrate on is preventing contamination
* dealing with isolated cells:
we are looking for vivo results, we explain invitro results as invivo because invivo give better results
* primary tissue culture (parental tissue):
- cells taken from cancer patients' liver, kidney, etc. are not pure (mixture of many cells)
- first cells are taken from the parental tissue, then we subculture it to become secondary tissue culture (pure) - one cell type

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- cell lines: - Adherant - non-adherant (insuspention)
- Adherant ⟶ in the bottom of the petridish of the flask they adhere (fix)
  non-adherant ⟶ they don't adhere and keep insuspention like lymphocytes

* manibulation of tissue culture must be preformed inside the tissue culture, while animal disection should be preformed outside to prevent contamination

* design of tissue culture lab:
- lab must be far away from microbial labs, animal house
- all windows and doors should be seeled with silicon to prevent entry of dust (bacteria)
- in the middle of the lab there's a UV light that should be turned on during night and off during the day to sterilize the environment (the lab's) during the night$

* main instruments for tissue culture:
- the lab must not be crowded, only main equipments should be kept inside like:

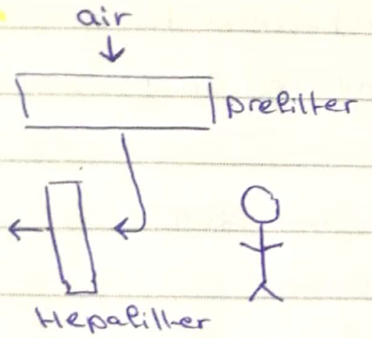
(1) laminar flow (hood)
(2) inverted microscope
(3) incubator
(4) centrifuge
(5) autoclave

* Laminar flow
- cabnet like a hood where we work inside
- this area must always be strile

- types of laminar flow:

① horizantal laminar flow
② vertical laminar flow:
   a. class I : no precations (not recommended)
   b+C. class II + III : prevent contamination of media and culture and more precise (recommended)

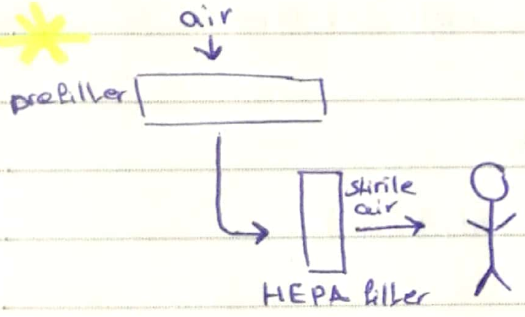
- vertical laminar flow (the one we use)



** for monoclonal Ab production it's better to work with vertical laminar flow because we are using cancerous cells (Myeloma cells)

remove anything greater than 0.3 μm, high effecient filter, most of bacteria and fugi will be removed

- horizantal laminar flow



** we use it if we're not working with something dangerous

* inverted microscope:
  for petridish (we can put flasks and petry dishes instead of slides)

* incubator (CO_2 incubator)
  concentration of CO_2 = 5% → if more it becomes toxic!
  37°C
  we fill the tray in the incubator with DW to get humid

environment → if humidity is less than wanted the cell will dry down

the water must be steril destilled water, otherwise it'll be a source of contamination, it also should be changed weekly (the water is boiled first)

$CO_2 + H_2O$ → buffer

* autoclave for glasswave at 121°c, 20 min, 15 pa.

we can't autoclave the media because the media contain glutamine which will be converted to glutamic acid if it was autoclaved

the media is sterilized by filteration

* disecting set is sterilized in oven:

wrap them with alamonium foil and put them in the oven at 160 c° 15-20 min

* pipets are also sterilized in the oven

pipets are put in metalic cylinder and in the oven (glass wave)

plastic wave are disposable

* centrifuge (refrigrated centrifuge)
* refrigerator:

media 4c°, f.c.s -20 c°, liquid nitrogen -196c° (cell survive for one year), short term storage (1 month) -70°--80°c°

* cleaning the lab:

- daily → mabbing the floor
- weekly → all surfaces, laminar flow cabinate with 70% alcohol, sink with disinfectant/detol, humidity tray 70% alcohol
- 3 months → dismantle the incubator

HYBRIDOMA LAB

LAB (4) Media

* RPMI - 1640 → trails

* RPMI (alternative media → MEME)

R: Rose P: Park M: Memorial I: Institution 16: scientist 40: trials

PH: 7.35 - 7.45

- powder:

contain ① sugar → ATP (energy)

② a.a → kreb's cycle (nutrients)

③ vitamin → methrathence mikerd

④ buffering system → control PH

⑤ PH indicator (phenol red) → difference in color according to PH ⑥ water (liquid)

* all of them help for survival for 24-48hr. *

- Liquid:

contain L-gluramine

* survive for 3 weeks *

bacteria fungi * phenol red *

0 yellow purple 14

6 7 8

orange ← Rose → blue

orange → high metabolism

turbidity → contamination → turbidity → color

if there's no ~~tur~~ turbidity it doesn't mean there's no contamination it might be cell death

* positivities: (powder vs liquid)

powder:

① ↑CO_2 (↑5%) ⇝ 1000 mL

② $NaHCO_3$ (s) buffer for PH 2g/1L ⇝ 0.2g/0.1L

③ HEPES: buffer 4.8 g/L ⇝ 0.48 g/L

* substances we add to the media:

① L-Gln : source of CO_2

※ ml = ※X (final volume) = 100ml = 1 ml

② penicillin+streptomycin

antibiotic ⟶ kill bacteria , liquid 50X, 100X

③ Amphotericin B 100X

②+③ ⟶ 100X $\frac{100}{100}$ = 1 ml

antimycotic ⟶ kill fungi

④ fetal calf serum: 100% f.c.s liquid

source of growth factor (hormones)

we will not add it today because f.c.s increase the chance of contamination

* f.c.s

90 ml media, 10ml f.c.s	80 ml media, 20ml f.c.s
10% V/V	20% V/V
growth and cell division	fussion and cloning

⑤ sodium ~~bacterium~~ bicarbonate buffer

$NaHCO_3$, solid, 2g/L ⇝ 0.2g/0.1L

⑥ HEPES: 4.77g/L ⟶ 0.477g/0.1L

* procedures

① weight powder from RPMI-1640, $NaHCO_3$, HEPES

② dissolve in 85 ml D.W

③ add liquid ~~from~~ form L-Gln, P+S+Amph B

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(4) adjust PH = 7.4 (7.35 - 7.45)

(5) complete volume up to 90 ml

under stirile conditions

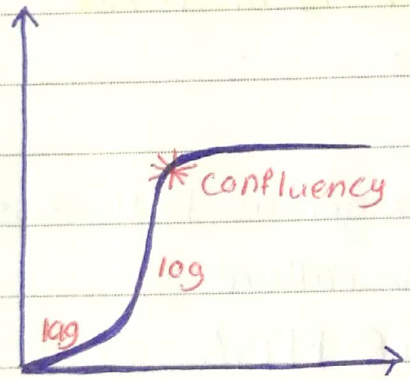
(6) sterilization by using filler until and put media in t.c.f

(7) put flask with media at incubator at 37°c for 24 hrs.

(8) transfer flask to refrigerator (4c°)

Hybridoma lec 5 Tissue culture contamination

* tissue culture:
1] Adherant
2] suspension
* substratant: monolayer of cells (adherant) that is confluent >80%.
* suspension: necrobiotic cells, cell line (is taken from primary tissue ex. liver tumor)
* cell → purify → culture/subculture 4-5 times
* ATCC: American Tissue Culture collection, has any type of cell line (in vitro))
* growth curve



- when there's confluency (too many cells) we subculture (putting new media)
- when the media is yellow → contamination / → confluency
- when morphology of the cells is deteriorating we subculture

* reasons of contamination:
1] Bacteria
- most common, the easiest to notice (yellow), the easiest to treat, divide very fast (over night to fill the flask)

- phenol red indicate the presence of bacteria (when contaminated phenol red turns yellow)
- when there's contamination we autoclave then discard
- add antibiotics to the culture:
  penicillin (anti gram +ve) + streptomycin (anti gram -ve)
- Gentamycin: we use it in rare cases and it's not recommended to prevent the resistancy, only liquid, only by injection, the last option antibiotic

2] yeast (fungi)

- less common, very fast growing, resistant to most antibiotics except Amphotericin and Nystatin, difficult to get rid off
- add Amphotericin B → 2.5 Mg/1 ml
  10 Mg/1 ml toxic

characteristic: fluffy balls

3] Mycoplasma

- not common but most dangerous to the tissue culture, kill the cells and lose the culture
- very hard to detect → PCR- ELISA
- it's a kind of bacteria without cell wall
- Antibiotics → Ampecillin ← no effect on mycoplasma
  streptomycin ←
  Tylosine → effective
- how it happens: a. cross contamination from other labs/cultures
  b. from personal hygiene (oral cavity)
- autoclave and discard
- has no signs

4 viruses

- not dangerous, has no effect on cells
- very small particles
- we can't get rid of it by filteration
- less than 1ML ⇝ pass through filter
- autoclave and discard
- cytopathogenic effect CPE
- dangerous to the person more than the culture

* media

- RPMI, DMEM

used for tissue culture to prepare monoclonal Ab

- powder	- liquid
industrial scale	for lab scale
long life shelf	steril
take small space	
cheaper	

- Gln (added fresh)

incubation for 1-4 weeks, 37°C

* PH (7,4) optimal

to regulate the PH:

a. bicarbonate buffer Na_2HCO_3, to prevent the acidic environment.

b. HEPs buffer, very efficient except infussion ⇝ toxic to the cell, will kill the cell

* fetal calf serum (f.c.s)

why we use calf? doesn't have IgG

5%, 10%	20%	0%
↓	↓	↓
through the work	fussion / cloning	washing cells

* B-mercapto ethanol

increase the production of Ab. biochemically effect

enhance the B-cell production of Ab

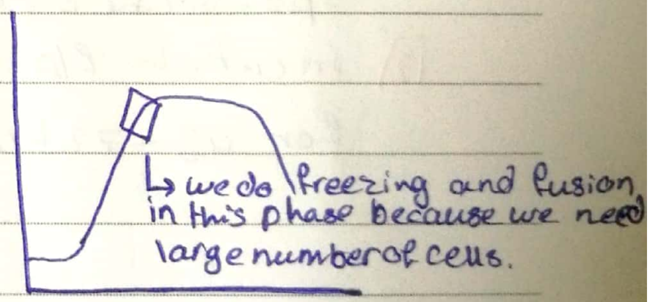
HYBRIDOMA LAB
LAB (5)

Maintanance of Myloma cells

* Myloma cells : abnormal lymphocyte.
  Sp.2
  mice origin

charechters of sp.2 : (1) cancerous (divide forever) as long as there are nutrients
(2) don't produce antibodies
(3) HGPRT enzyme (-ve) → blocked
" this enzyme is for pentose pathway "

* Thawing:
transfering cells from -196°
(liquid nitrogen) solid form to
37°C liquid form
caryotub (freezing tube) is a
tube that can tolerate extremly
low temperature



DMSO : a very toxic substance that we add to freeze substances.

* prosedure:
1] remove caryotubes from liquid nitroge jar and immidielly thaw them at water bath for 2-3 min. with constant mixing
W.b
2] under strile conditions (binzel burner) wash the tube with alcohol

b.b
3] transfer cells + media to sterile centrifuge tube
b.b by using a stirile syringe
4] add 5 mL 50% F.C.S (media + 50 F.CS) drop by drop
over 2 min.
5] centrifuge at 2500 rpm / 5 min
b.b
6] At sterile petry dish decant supernatent
b.b
7] Add 2-5 ml washing media (no ~~[illegible]~~ F.C.S) to pellet
immediately
8] centrifuge at 2500 rpm / 5 min.
b.b
9] decant supernatant in sterile petridish
b.b
10] add 8-10 mL 10% F.C.S to pellet and mix
b.b
11] transfer cells + media to sterile tissue culture
flask and Label
sp2, 10% F.C.S, 6-3-2019, names, group ✱, section 2
12] incubate flask with cells at 5%, 37° incubator
for 48-72 hrs.

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Hybridoma Lec 6 Myeloma cells

* S-cells → pro-B → pre-B → immature B → mature B → plasma cell (multible myeloma)
* Myeloma → cancerous plasma cell
multible myeloma → cancer patient
* Myeloma + B-cell (that produce Ab) fussion
they keep dividing   injected mouse
* characterastics of Myeloma cells
1] HGPRT
2] none Ab producing
3] none Ab secreting
* why we don't use B cells alone in tissue culture ?
because it will die spontanously because it's not cancer cell
* the cell produced from fussion → Hybridoma cell
* Myeloma cell and B-cell look alike under microscope
* DNA synthesis → denovo (simple atoms H, C, N, O)
→ salvage pathway (to rescue)
• when we use AMP for synthesis it's considered recycling
• HGPRT → recycles
when HGPRT⁻ it means there's a diffeciency in that enzyme which makes the DNA unable to do salvage pathway (a mutation by Azaguanine or Thioguanine)

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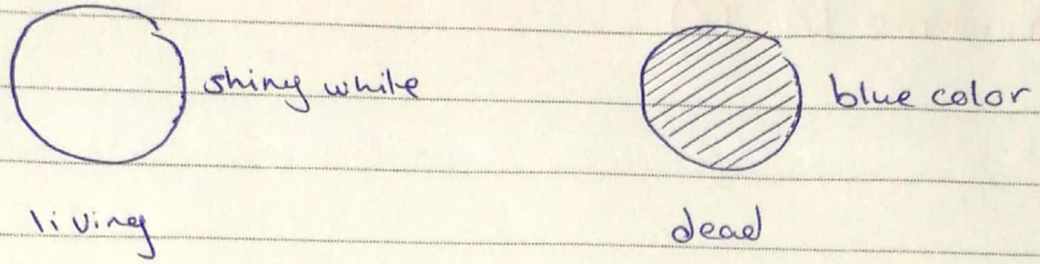
* B-cells + Myeloma cells $\xrightarrow{HAT}$ B-B (die)
  → M-M (don't divide because we inhibited the pathway so they'll die)
  → M
  → B (die)
  → B-M (hybridoma cell) (salvage pathway stays active)

HAT: • H → hypoxanthin
• A → Aminopterine (inhibitor for one of the steps of denovo)
• T → Thymidine (Nitrogen base + sugar)

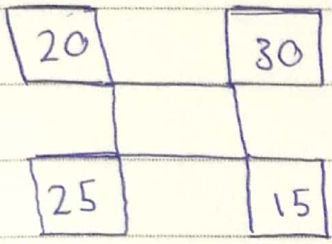
* when we buy Myeloma it's HGPRT⁻ and this what inhibits the pathway (second one) in M-M (the salvage pathway)

* Maintainance
• 50 ml TC flask
• density $10^5 - 5\times10^6$
• every 2-3 months grow myeloma cells in Azaguanine which inhibit DNA synthesis, so the HGRPT + Myeloma cell will die and the HGRPT⁻ myeloma cells will survive
• criteria of Myeloma cell for fussion
log phase → high growth (last division)
density → 10^6 cell
viability → >95% living cell

* Viability test
Trypan blue (stain protein)
selective permeability for stain



- Counting by hematocytometer



through counting we shouldn't use f.e.s

Average : $20 \times 10^5 = 2 \times 10^5$ /ml

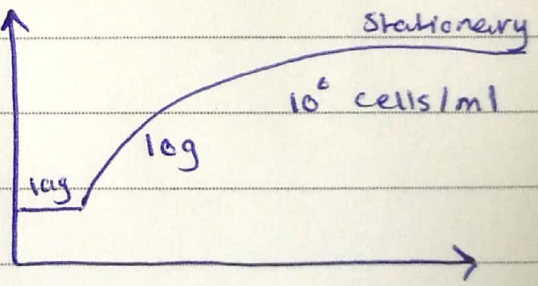
* characteristic of hybridoma cell:

1] ~~...~~ Immortal

2] produce Abs

* counting for sp^2 (abnormal lymphocytes) by hemocytometer:

- freezing of fussion at late log phase, number of sp^2 1×10^6 cells/ml when we can do freezing and fussion 1:10 fussion



- counting by using stain: Trypan blue / nigrosin

* Trypan blue (viable stain)

viable		dead
round		wrinkled
shiny		blue

** should be counted within 5-10 min, after that all cells become dead (blue) **